

Summary

Software Engineer and AI Researcher with 3+ years of experience in deep learning and backend development. Focused on optimizing vision-language models like CLIP for video anomaly detection using prompt learning, achieving a 15% efficiency gain. Improved media search performance by 83% through Java-based microservices. Published researcher with practical experience deploying AI into production environments.

Education

Rhineland Pfalz Technical University(RPTU)

GPA : 2.1

M.Sc. Computer Science

(2023 - Present)

Bharati Vidyapeeth's College of Engineering, GGSIPU, New Delhi, India

GPA : 8.16/10

Bachelor of Technology, Computer Science and Engineering

(2016 - 2020)

Technical Skills

Programming Languages – Python, Java, C++

Frameworks and Tools – Pytorch, Tensorflow, Transformers, Huggingface.

ML Skills – Computer Vision, NLP, Transformers, Deep Learning, Diffusion models, Prompt Learning, Fine-tuning, LLM, MCP.

Databases – SQL(SAP HANA), MySQL, NoSQL(MongoDB).

Cloud Technologies – GCP-(AppEngine, CloudRun, Kubernetes, Datastore), AWS Azure

Publications

Driver distraction detection using capsule network: Developed a **capsule network** architecture that achieved 95% driver distraction detection accuracy under real-world lighting conditions. ([Link](#))

IF - 5.6

Distracted driver detection using **compressed energy efficient** convolutional neural network: Proposed a compressed, energy-efficient CNN for distracted driver detection, balancing accuracy and resource use([Link](#))

IF - 1.96

Professional Experience

Research Assistant, RPTU

(May 2024 - Present)

Video Anomaly detection

Customized **vision language model**, CLIP with prompt learning mechanism for UCF-Crime anomaly detection, improving temporal localization by 15% and cutting inference time by 80%.

Engineer, Nagarro Pvt Ltd

(Oct 2020 – April 2023)

Media enhancement application: Developed a Java-based HD/4K video **transcoding** system with built-in object and scene detection, boosting throughput by 34% and cutting memory usage by 11% on Android/iOS (DASH/HLS).

Spend analytics and customized offers for a banking client: Built a dynamic spend analytics dashboard processing users' monthly transactions with a **semantic embedding-based personalization engine**, increasing offer engagement 25%.

Feed Generator system for a media publishing client: Developed a **GraphQL Java-based** solution with **Springboot** and deployed it on Kubernetes from ~20 sources to generate customized feed for different client partners

Cross Product search for a media publishing client: Improved search speed by 83% for a major media client by designing typo-tolerant **Elasticsearch** pipelines using **Kafka** and **SpringBoot**.

Internships

Fake news detection

(Aug 2019 – Oct 2020) I.I.T Delhi

Achieved 95% accuracy detecting morphed images with a **Siamese network** using **SIFT** and **hashing**; cut false positives 12% via multi-feature fusion. Built scalable labeling with **Google Vision API**.

Complaints Prediction

(June 2019 – June 2019) HPCL

Analyzed and applied **Transformer**-based model to forecast complaint volume with 92% accuracy; implemented BERT summarization, cutting review time by 60%.

Side Projects

Synthetic Image generation model

Applied **Diffusion** model on public images along with customized input text to generate new images and made the dataset public